

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Mern Stack

Module - I

Practical – II

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Class:TYIT	Batch:2
Date of Assignment:30/06/26	Date/Time of Submission: 30/06/26

2. Core JavaScript Concepts for MERN

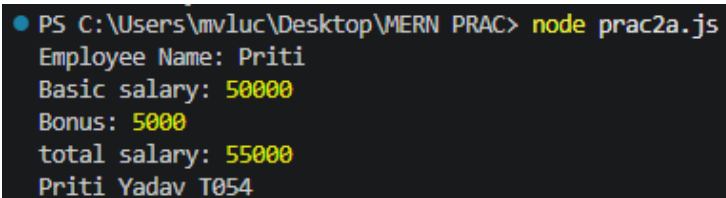
a. Create functions and call them with parameters.

CODE:-

For calculateSalary

```
function calculateSalary(name,basicSalary){  
  let bonus=5000;  
  let totalSalary=basicSalary+bonus;  
  console.log("Employee Name:",name);  
  console.log("Basic salary:",basicSalary);  
  console.log("Bonus:",bonus);  
  console.log("total salary:",totalSalary);  
}  
calculateSalary("Priti",50000);  
console.log("Priti Yadav T054");
```

OUTPUT:-



```
PS C:\Users\mvluc\Desktop\MERN PRAC> node prac2a.js  
Employee Name: Priti  
Basic salary: 50000  
Bonus: 5000  
total salary: 55000  
Priti Yadav T054
```

For Generating bill with 18% GST.

CODE:-

```
function genBill(Product,Quantity,Price){  
  let total=Quantity*Price;  
  let bill=total+(total*0.18);  
  console.log("Product:",Product);  
  console.log("Quantity:",Quantity);  
  console.log("Price per Item:",Price);  
  console.log("total Bill:",bill);  
}  
genBill("Fridge",2,50000);  
console.log("Priti Yadav T054");
```

OUTPUT:-

```
PS C:\Users\mv1uc\Desktop\MERN PRAC> node practical2A.js
Product: Fridge
Quantity: 2
Price per Item: 50000
total Bill: 118000
Priti Yadav T054
```

b. Work with arrays and objects in JavaScript.**CODE:-**

```
//Array
let fruits=["Mongo","Banana","Apple","Cherry"];
//Object
let student={
  name:"Priti",
  age:19,
  city:"Kolkata"
};
console.log("Fruits:",fruits);
console.log("Second Fruits:",fruits[1]);
console.log("Student Name:",student.name);
console.log("Student Age:",student.age);
console.log("Student city:",student.city);
console.log("Priti Yadav T054");
```

OUTPUT:-

```
PS C:\Users\mv1uc\Desktop\MERN PRAC> node prac2b.js
Fruits: [ 'Mongo', 'Banana', 'Apple', 'Cherry' ]
Second Fruits: Banana
Student Name: Priti
Student Age: 19
Student city: Kolkata
Priti Yadav T054
```

c. Demonstrate use of arrow functions.**CODE:-**

```
let add=(a,b)=>{
  return a+b
};
let sub=(a,b)=>{
  return a-b
};
let mul=(a,b)=>{
  return a*b
};
let div=(a,b)=>{
  return a/b
};
console.log("Addition:",add(10,10));
console.log("Subtraction:",(22,19));
console.log("Multiplication:",(10,10));
console.log("Division:",(15,5));
console.log("Priti Yadav T054");
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\MERN PRAC> node prac2c.js
Addition: 20
Subtraction: 19
Multiplication: 10
Division: 5
Priti Yadav T054
```

d. Write a program using array methods (map, filter, reduce).**CODE:-**

```
let numbers=[10,20,30,40,50];
//map()
let doubled=numbers.map(num =>num*2);
//filter
let greater25=numbers.filter(num=>num>25);
//reduce
let total=numbers.reduce((sum,num)=>sum+num,0);
console.log("Original Array:",numbers);
console.log("Map:",doubled);
console.log("Filter:",greater25);
console.log("Reduce:",total);
console.log("Priti Yadav T054");
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\MERN PRAC> node pracd.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
Priti Yadav T054
```

e. Create a program that manipulates objects and displays output.**CODE:-**

```
//Creating an object
let student={
  rollNo:54,
  name:"Priti",
  marks:83
};
console.log("Original Object");
console.log(student);
//2.Accessing Properties
console.log("\nAccessing Properties:");
console.log("Name:",student.name);
console.log("Marks:",student.marks);
//updating properties
student.marks=90;
console.log("\nAfter Updating Marks:");
console.log(student);
//Adding a new Property
student.marks=90;
console.log("\nAfter updating marks:");
console.log(student);
```

```
//Deleting a Property
delete student.rollNo;
console.log("\nAfter Deleting Roll Number:");
console.log(student);
console.log("Priti Yadav T054");
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\MERN PRAC> node prac2e.js
Original Object
{ rollNo: 54, name: 'Priti', marks: 83 }

Accessing Properties:
Name: Priti
Marks: 83

After Updating Marks:
{ rollNo: 54, name: 'Priti', marks: 90 }

After updating marks:
{ rollNo: 54, name: 'Priti', marks: 90 }

After Deleting Roll Number:
{ name: 'Priti', marks: 90 }
Priti Yadav T054
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

CODE:-

```
//Array of object
let students=[
  {
    rollNo:54,
    name:"Priti",
    marks:83
  },
  {
    rollNo:32,
    name:"khushiii",
    marks:90
  },
  {
    rollNo:7,
    name:"subuu",
    marks:92
  },
  {
    rollNo:31,
    name:"ishuu",
    marks:98
  }
];
//function display
function display(s)
{
```

```
console.log("Student details");
console.log("-----");
for(let stud of s)
{
  console.log(`Roll No:${stud.rollNo}`);
  console.log(`Name:${stud.name}`);
  console.log(`Marks:${stud.marks}`);
  console.log("-----");
}
}
//function call
display(students);
console.log("Priti Yadav T054");
```

OUTPUT:-

```
PS C:\Users\mvluc\Desktop\MERN PRAC> node prac2f.js
Student details
-----
Roll No:54
Name:Priti
Marks:83
-----
Roll No:32
Name:khushiii
Marks:90
-----
Roll No:7
Name:subuu
Marks:92
-----
Roll No:31
Name:ishuu
Marks:98
-----
Priti Yadav T054
```